PROFESSIONAL SMART EYEWEAR

Engineering Reference

Twelve findings with stated conditions.
Fifteen chapters linking architecture to evidence.

Every performance claim needs a workload, a configuration and a measurement.

Scoped findings · Calculation methods · Validation evidence
Standards references current as of 8 September 2026

Supersedes Engineering Certified Smart Eyewear, Edition 1.1.

One technical position. A traceable revision.

Edition 1.2 supersedes Engineering Certified Smart Eyewear, Edition 1.1, for technical guidance.

This revision retains the engineering questions behind the earlier reference and corrects conclusions that exceeded their evidence. It separates public-source facts, explicit calculations and engineering recommendations. None of the numerical examples is represented as measured performance of a SmartXY product.

TOPIC	EDITION 1.2 POSITION
Impact / ingress	Interface conflicts merit testing; architecture features alone do not establish mutual exclusion.
Thermal power	Publish the heat-transfer coefficient and model assumptions with any estimate.
Market references	Use ANSI/ISEA Z87.1-2025, current CSA Z94.3:26 pointers and the applicable EU route.
MFi and platform reuse	Technology and retained scope determine applicability, dependencies and possible savings.
Costs and contracts	Use scope-based models; prior customer-level figures and negotiated terms are excluded.

Document roles: this reference supplies technical findings and methods. The separate Buyer Decision Guide, Edition 1.4, provides evaluation questions and integration tools. Its carousel and launch kit support communication. All use the same substantive positions.

Publication control: public releases must exclude customer-confidential material and use customer-identifying designs, marks or project data only with documented permission. This technical edition contains no product renderings, customer case study, project calendar or customer quotation.

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READER MAP

Find the decision and its evidence.

Read each finding with its scope. Use the corresponding chapter before applying it.

CHAPTER / REFERENCE	PAGE
Twelve scoped findings, F-01 to F-12	**4-5**
01 · Protection, markets and section 1.4 Europe	**6**
02 · Impact, ingress and service interfaces	**7**
03 · Lens and frame architecture	**8**
04 · Prescription compatibility and fit	**9**
05 · Thermal requirements and calculated model	**10-11**
06 · Speech interaction and validation	**12**
07 · Capture geometry and OCR acceptance	**13-14**
08 · Energy, compute and integration	**15-16**
09 · Lifecycle economics	**17**
10 · Cleaning and environmental qualification	**18**
11 · Assessment, Europe and the dependency calendar	**19-21**
12 · Development gates and production evidence	**22**
13 · Reuse and scope-based cost model	**23-24**
14 · IP, source access and commercial controls	**25**
15 · Pre-commitment questions	**26**
Canonical editorial principles / public references	**27 / 28-29**

Evidence labels: a source-backed fact reports what the cited authority says; a calculated example follows from stated inputs; an engineering synthesis proposes a defensible decision or verification method. These labels prevent a recommendation from becoming an invented mandatory clause.

Twelve findings. Their conditions included.

Reference these statements together with the scope immediately below them.

F-01 **Ports and service interfaces couple impact, ingress and maintenance requirements. Their coexistence does not establish that high impact and IP68 are impossible.**

Engineering synthesis. Evaluate the assembled enclosure, tolerances and credible exposure sequences; do not select IP54 by default.

F-02 **A prescription carrier changes the optical and mechanical configuration of a protector.**

Engineering synthesis. Evaluate retention, eye clearance, fitting and representative prescription variants through the applicable route.

F-03 **Electrical safety, contact comfort and sustained function are separate acceptance requirements.**

Engineering synthesis. A temperature that satisfies one requirement does not, by itself, establish the other two.

F-04 **A stated convection-only example gives 0.51-1.44 W when h = 8 W/m²·K, area = 0.008-0.012 m² and temperature rise = 8-15 K.**

Calculated example. This is heat transfer under assumed boundary conditions, not a universal device-power, safety or comfort limit.

F-05 **Under the specified ideal camera geometry, a 4 mm horizontal feature occupies about 18 pixels.**

Calculated example. 3,840 horizontal pixels, 80° horizontal FOV, 500 mm distance and a flat target perpendicular to the optical axis; no OCR pass threshold is implied.

F-06 **The specified 0.95 Wh energy example supports about 86 mW average over eight hours, or about 26 minutes at a constant 1.6 W.**

Calculated example. Assumes 85% conversion efficiency and a 15% reserve. Actual local-AI feasibility depends on the workload.

Twelve findings. Their conditions included.

Reference these statements together with the scope immediately below them.

F-07 **Replacing an optical consumable can preserve functioning electronics; the economic benefit depends on the delivered service process.**

Engineering/economic model. Compare replacement, authorized repair and front-module exchange using frequency, cost and downtime inputs.

F-08 **Cleaning compatibility is specific to materials, assembly stress and the cleaning protocol.**

Engineering synthesis. Coupon screening cannot establish finished-assembly life; service-cycle claims require a usage model.

F-09 **Assessment can overlap design validation when the assessment body accepts the samples and the configuration is controlled.**

Conditional planning principle. Changes require impact assessment; retest scope is neither automatically zero nor automatically the entire program.

F-10 **A usable schedule connects work to sample readiness and the availability of the actual suppliers and laboratories.**

Planning principle. Use confirmed queues, shutdowns and recovery dates rather than a universal holiday allowance.

F-11 **Platform reuse has three distinct assets: technology, applicable evidence and permission to use both.**

Engineering/commercial principle. Benefits depend on retained assets, change scope, supplier continuity and assessment acceptance.

F-12 **Development cost and schedule must be derived from the selected starting point and remaining scope.**

Commercial method. Customer-specific quotes, fixed margins and negotiated payment terms do not establish category-wide benchmarks.

Protection starts with a defined intended use.

Source-backed facts [1-5] and engineering interpretation are identified separately.

1.1 Define the complete product

Name the hazards, users, markets, protector configuration and claims. Professional eyewear can be used without an eye-protection requirement, with assessed compatible protection, or as an integrated protector. Added electronics, openings, masses and accessories need evaluation within the applicable configuration.

1.2 United States

ISEA announced ANSI/ISEA Z87.1-2025 on 8 January 2026. Publication and regulatory adoption are different matters. Assess workplace requirements under OSHA 29 CFR 1910.133 and the relevant conformity route; avoid the unsupported phrase OSHA-certified glasses. [1,2]

1.3 Canada

CSA lists Z94.3:26 as active and its Z94.3:20 (R2025) listing as withdrawn. Confirm the edition applicable to the jurisdiction, purchaser and assessment program. An assessment for one market does not automatically establish conformity for another. [3]

1.4 Europe: current planning basis

EN 166:2001 lost its EU harmonised-reference status on 11 November 2025. Use the applicable PPE conformity route and verify the precise EN ISO 16321-1 reference and transition provisions at assessment time. The publisher's edition including A1:2025 and an Official Journal reference are distinct facts. Legacy F/B/A tables are not a current design brief for every new spectacle. [4,5]

A target, test report, declaration and third-party certification are different evidence. Match claims to the actual route and configuration. Impact protection alone establishes no general suitability for splash, welding, lasers, electrical arc or explosive atmospheres.

Impact, ingress and service interact.

Conditional engineering position: combined requirements need system-level validation.

F-01 **Ports and service interfaces couple impact, ingress and maintenance requirements. Their coexistence does not establish that high impact and IP68 are impossible.**

Engineering synthesis. Evaluate the assembled enclosure, tolerances and credible exposure sequences; do not select IP54 by default.

Acoustic openings, a serviceable battery and a replaceable lens create interfaces, but they do not specify enclosure topology, sealing technology or the mechanical load path. A lens interface may sit outside the electronics enclosure. These features cannot, by themselves, prove that IP68 and high-impact performance are mutually exclusive.

INTERFACE	COUPLED QUESTIONS
Lens retention	How does it behave under loading, assembly variation and repeated replacement? Is it actually part of the electronics enclosure?
Acoustic openings	What changes when membranes are wet, dirty, aged or assembled with production variation?
Battery and charging	How are contact wear, corrosion, debris, incorrect insertion and electrical faults controlled?
Housing joints	How do material choice, bond preparation, residual stress and movement affect integrity?

IEC 60529 classifies ingress protection. Select the target from intended exposure rather than treating IP54 as universally correct. Agree the assembled states, tolerances, ageing and credible test sequences that support the claim. [6]

Examples of supplemental sequences include servicing followed by ingress evaluation, or cleaning and thermal ageing followed by mechanical assessment. Keep these separate from mandatory conformity tests. Define inspection and retirement after impact; electronics continuing to run is not evidence of continued protective suitability.

Geometry connects coverage, optics and service.

Engineering synthesis: compare the complete configuration rather than one lens parameter.

OPTION	POTENTIAL VALUE	WORK TO RESOLVE
Single shield	Continuous front coverage.	Optical quality, retention, ventilation and insert clearance.
Dual lens	Familiar form and prescription supply options.	Side coverage, bridge stiffness, lens retention and electronics packaging.
Over-glasses	Accommodation of an existing prescription pair.	Stable combined fit, visibility, fogging and compatibility with other PPE.

Coverage is three-dimensional

Base curve alone cannot define protection or fit. Evaluate wrap, frame dimensions, lens shape, side geometry and position on the face. Allow space for eyelashes, prescription components and the expected movement of the assembly.

Give the camera an optical specification

A separate window and a view through the protective lens create different imaging and structural questions. Compare focus, distortion, reflections, transmission, contamination and image quality in the assembled configuration. Any aperture also needs a mechanical review.

Make replacement a controlled operation

Define compatible parts, positive seating, clear assembly feedback, instructions and traceability. Establish which lens and serviced variants are covered by the assessment. A replaceable component still needs a controlled replacement process.

Prescription components are part of the system.

Scope: the optical and mechanical configuration actually offered to the user.

F-02 **A prescription carrier changes the optical and mechanical configuration of a protector.**

Engineering synthesis. Evaluate retention, eye clearance, fitting and representative prescription variants through the applicable route.

APPROACH	EVIDENCE TO ESTABLISH
Prescription protective lenses	Supported prescriptions, materials, glazing route, markings, ordering and replacement.
Prescription insert	Complete-system optics, retention, eye clearance, supported variants and fitting process.
Over-glasses arrangement	Stable position with intended frames, visibility, comfort and other protective equipment.

OSHA describes both prescription-integrated protection and protection worn over prescription lenses without disturbing proper position. Neither prescription architecture is universally required for all tasks. [2]

Measure vertex distance, wrap, tilt, pupillary distance, thickness and the prescription envelope. An insert can change vertex distance in either direction. Have a qualified optical partner establish compensation and dispensing; a fixed millimeter offset or universal diopter threshold is not a substitute.

Evaluate retention and clearance under relevant loads with representative prescription variants. A detachable carrier is not inherently safer. Include varied faces, movement and other protective equipment in fit trials; record pressure, slip, visual complaints and task performance.

The release output is a supported fit and prescription envelope with ordering, fitting, replacement and inspection instructions. The limits must be usable by the channel before an order is accepted.

Thermal acceptance has three dimensions.

Separate the evidence for safety, comfort and the ability to continue the task.

F-03 **Electrical safety, contact comfort and sustained function are separate acceptance requirements.**

Engineering synthesis. A temperature that satisfies one requirement does not, by itself, establish the other two.

REQUIREMENT	MEASURE
Safety	Applicable limits, component and battery conditions, foreseeable faults and required protection behavior.
Comfort	Contact-point temperature, pressure, hotspot location and user tolerance over the intended session.
Continuity	Capture, speech, radio and compute behavior as the device warms and thermal controls activate.

Measure the final arrangement at relevant ambient extremes, solar and airflow conditions, activity profiles and radio states. Record skin-contact hotspots, internal component temperatures and task behavior. Include ageing and production variation in the margin plan.

A spreader redistributes heat and may warm a larger contact area. An insulating layer can reduce skin heat flow while increasing internal temperatures. Validate the geometry and electrical and mechanical integration rather than treating a material stack as a universal recipe.

Agree warnings, reduced processing, session limits and safe shutdown where appropriate. Verify that electronic limiting leaves the passive protective function intact for its assessed use and communicates status clearly. A single touch-temperature number cannot establish all-day comfort for every wearer.

A watt estimate needs a heat-transfer model.

Convection-only screening: Qconv = h × A × (Ts - Ta).

Assume a uniform exposed surface, a convection coefficient h = 8 W/m²·K, area A = 80-120 cm² = 0.008-0.012 m² and temperature rise ΔT = 8-15 K. The table applies the extremes of those assumed ranges; it is not measured device performance.

SCENARIO	INPUTS	CONVECTIVE HEAT
Low example	8 × 0.008 × 8	0.512 W ≈ 0.51 W
Mid example	8 × 0.010 × 10	0.800 W
High example	8 × 0.012 × 15	1.440 W ≈ 1.44 W

Within these assumptions, estimated convective heat transfer spans 0.51-1.44 W. The result does not define a universal one-watt eyewear ceiling.

What this model leaves out

Radiation, conduction into skin and adjacent structures, nonuniform temperatures, changing airflow and solar input are not solved here. Component power and heat rejected from the exposed surface are not interchangeable unless the complete energy balance is established. The coefficient is an assumption, not a value fixed by the phrase still air.

Make the estimate useful

Define the system boundary and proposed load map. Measure power and local temperatures on representative builds, identify dominant heat paths, and calibrate a suitable model. Verify the worst intended operating cases and limiting behavior. Increasing ΔT in this equation does not authorize an unsafe or uncomfortable surface temperature.

Speech performance belongs to the whole exchange.

Engineering synthesis: input detection, wearer selection, action, feedback and recovery.

STAGE	FAILURE TO LOOK FOR
Wake and detection	Clipped word beginnings or endings; false activation from tools, movement or nearby talkers.
Wearer selection	Confusion between wearer and bystander speech across fit, movement and noise.
Echo and playback	Failures during interruption, profile changes or simultaneous assistant and call audio.
Action and feedback	Wrong actions, misunderstood confirmations, response delay and poor recovery.

Build a reproducible corpus covering relevant noise spectra, impulses, wind, talker positions, frame movement and protective equipment. Record acoustic conditions and microphone placement. A microphone count or a bare dB figure is not an end-to-end performance specification.

Pre-roll helps preserve the start of an utterance; end-point policy trades completeness for delay. Include this delay in the total response budget. Echo cancellation needs an accurate, synchronized playback reference, but a hardware post-amplifier feedback path is not universally mandatory.

Test every intended call, media and assistant route, including interruption and simultaneous playback. Vibration sensing and beamforming can support wearer selection, but contact, fit, movement and processing affect both.

Report task completion, false activation, false actions, response-time distribution and recovery. Evaluate feedback with the required hearing protection. Use controlled recordings and fixtures where possible and an appropriate exposure-control plan for human trials.

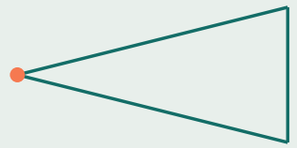
Pixels on the target follow the projection model.

Ideal rectilinear camera viewing a flat target perpendicular to the optical axis.

IDEAL RECTILINEAR GEOMETRY

3,840 px / 839 mm

≈ 4.6 pixels per millimeter



500 mm to target

Schematic; not to scale

Let horizontal image width be N = 3,840 pixels, horizontal FOV = 80° and axial target distance D = 500 mm. Scene width W = 2D × tan(HFOV/2) = 839.10 mm. Therefore N/W = 4.576 pixels/mm, or approximately 4.6 pixels/mm across the ideal frontoparallel flat target. A horizontal 4 mm feature occupies 18.31 pixels.

For this ideal model, ximage = f × X/D, so the pixel-to-length scale on that plane is uniform. The phrase near the center is unnecessary under these assumptions. Real distortion, rectification or resampling, target tilt, curvature and depth variation can change local scale.

The calculation addresses geometry, not optical transfer, blur, noise, exposure, focus, compression or recognition. It establishes no universal minimum number of pixels for OCR. Wider FOV increases context; narrower FOV can place more pixels on a fixed target at a fixed distance.

Use actual targets before freezing camera and frame geometry. Test through the final lens or window. Digital cropping cannot recreate missing detail, and electronic stabilization cannot by itself remove blur accumulated during exposure.

Character accuracy can hide identifier failures.

Calculated illustration and a proposed validation method, not measured results.

If each character were independently correct with probability 0.95, a 17-character string would be entirely correct with probability 0.95^17 ≈ 0.418, or 41.8%.

The independence assumption is illustrative and often unrealistic. Its purpose is to show why average character accuracy cannot substitute for the exact-field outcome required by an asset, part or serial-number workflow.

MEASURE	REPORT WITH IT
Exact-field success	Correct complete fields / attempted fields, with task and sample definitions.
False acceptance	Wrong outputs accepted as correct / explicitly defined evaluation population.
Retries and time	Retry count, abandonment and time to an accepted result, including recovery.
Conditions	Target size, distance, illumination, reflectivity, motion, focus, build and software versions.
Production correlation	Focus, alignment and window tests correlated with the relevant task outcomes.

Set acceptance thresholds from the consequence of error and the intended task; do not publish an arbitrary percent as a category requirement. Stratify the evaluation by important failure conditions and report sample size and uncertainty.

Use confirmation or escalation where consequences warrant it. Preserve source evidence where policy permits. A plausible answer is not proof of a correct identifier.

Runtime is the quotient of usable energy and load.

Calculated screening method: t = Eusable / Paverage.

0.686 Wh	86 mW	26 min
usable energy	average for 8 hours	at a sustained 1.6 W

Assume 0.95 Wh nominal energy, 85% conversion efficiency and a 15% reserve. Eusable = 0.95 × 0.85 × 0.85 = 0.686375 Wh. Over eight hours, the allowable average is 85.797 mW, approximately 86 mW. At a constant 1.6 W, t = 0.42898 h = 25.74 minutes, approximately 26 minutes.

These inputs are hypothetical. Reserve convention, conversion loss and measured battery capacity must be defined to avoid double counting. Replace average-load assumptions with measured state power, state duration and event energy.

MODEL LAYER	MEASURE OR DEFINE
Device states	Standby, sensing, capture, encoding, transfer, playback and inference; include transitions and retries.
Host boundary	Power consumed by the phone or gateway required for the workflow. Offload moves work; it does not eliminate system energy.
Replenishment	Ageing and temperature margins, charging opportunities, exchange workflow and shift-start readiness.

Low-power local inference and long runtime can coexist. Feasibility depends on workload, silicon, duty cycle and the complete system. A calculation for sustained high-power inference does not prove that every all-day on-device AI workload is impossible.

Integration is a configuration-specific result.

Source-backed MFi scope [7]; capability and offline validation are engineering recommendations.

BOUNDARY	MAKE IT TESTABLE
Microphone	Raw or processed audio, format, sample rate, access route, cloud dependency and allowed concurrency.
Camera	Still capture, recording, preview or streaming; resolution, latency, transport and thermal limits.
Controls and sensors	Events, timestamps, rates, mapping, buffering and behavior during disconnect.
Models and firmware	Permitted deployment targets, memory and compute limits, signing, updates and recovery ownership.
Host and transport	Supported OS versions, permissions, background operation, pairing, reconnection and throughput.

For each function, report available and evidenced, requires development, under evaluation or unsupported. Identify hardware, firmware, SDK, host OS, concurrent activities and test evidence. API access, selected source access and ownership of the stack are separate deliverables.

Apple determines MFi scope from licensed technology, not the presence of an iPhone. Accessories using only BLE or supported standard Bluetooth profiles without covered technology are generally outside MFi, subject to listed exceptions. No universal three-to-six-month MFi critical path applies to every smart-eyewear design. [7]

Offline capture, offline inference and isolated-network operation are different requirements. Validate activation, authentication, licensing, storage, updates, recovery, reassignment and retirement under the intended restrictions. Define data access, retention and deletion and test actual host behavior.

Serviceability changes the cost of staying useful.

Conditional economic model, with input values from the intended deployment.

F-07 **Replacing an optical consumable can preserve functioning electronics; the economic benefit depends on the delivered service process.**

Engineering/economic model. Compare replacement, authorized repair and front-module exchange using frequency, cost and downtime inputs.

COST LAYER	INCLUDE
Acquire and deploy	Hardware, accessories, integration, training, prescription fitting and initial spares.
Operate	Software or AI usage, connectivity, charging, cleaning, inspection and fleet administration.
Service	Lenses, batteries, labor, shipping, repair turnaround and loaner devices.
Retire and recover	Loss, damage, secure data removal, end-of-life handling and applicable recovery obligations.

Annual service cost = event frequency × delivered event cost + downtime cost + support overhead.

Delivered event cost includes the part, labor, logistics, inspection and any required requalification. State which overheads are already included to avoid double counting. Estimate replacement frequency and downtime from the intended use or a representative pilot.

Compare field replacement, authorized factory repair and front-module exchange. A bonded lens does not automatically make every business model unviable; a cheap spare does not automatically produce the lowest total cost.

Control compatible parts, service instructions, traceability and retirement criteria. Treat changes to lens, coating, material and retention through configuration control. Keep unproven productivity benefits out of guaranteed savings.

Cleaning qualification needs the assembled product.

Scope the chemistry, stress and exposure model before making service-life claims.

F-08 **Cleaning compatibility is specific to materials, assembly stress and the cleaning protocol.**

Engineering synthesis. Coupon screening cannot establish finished-assembly life; service-cycle claims require a usage model.

DEFINE	CHECK AFTER EXPOSURE
Agent and method	Formulation, dilution, wiping, dwell time, rinse, drying and required equipment.
Assembly condition	Stress, material and coating lots, prior ageing and temperature.
Exposure model	Cycles from the use pattern and service interval; inspection and replacement points.
Optics and structure	Haze, transmission, distortion, cracks, retention, bonds and readable markings.
Electronics and audio	Charging, corrosion, ingress paths, acoustic response and wet-state recovery.

Formulation, material grade, coating, bond preparation, residual stress, temperature and dwell time can change the result. There is no evidence here that every workplace uses IPA, that cleaning causes most failures or that every protector must tolerate every solvent.

Screen materials with coupons, then qualify assembled parts. Where risk justifies it, evaluate mechanical or ingress behavior after cleaning and ageing. Distinguish supplemental engineering sequences from formal conformity requirements.

Assess anti-fog behavior after the intended cleaning protocol. A laboratory cycle count becomes a field-life estimate only through a stated use model. Release approved methods, prohibited use, inspection intervals and retirement criteria that match the evidence.

Concurrent assessment is a controlled risk decision.

11.1 Sample readiness and evidence ownership.

F-09 **Assessment can overlap design validation when the assessment body accepts the samples and the configuration is controlled.**

Conditional planning principle. Changes require impact assessment; retest scope is neither automatically zero nor automatically the entire program.

WORKSTREAM	EARLY DELIVERABLE
Protection	Markets, hazards, variants, sample matrix and the assessed configuration. [1-5]
Electrical / radio	Applicable safety, EMC, radio and head-worn RF-exposure evaluations.
Battery / transport	Cell, pack and finished-product evidence and documentation for the shipped configuration. [8]
Wireless / platform	Bluetooth qualification responsibility and any actual MFi technology scope. [7,9]

Agree representative sample requirements with the responsible body. First production tooling may be suitable in a program, but DVT1 is not a universal submission point. Document configuration stability and who owns changed samples, additional tests and retest budget.

Assess each change against affected evidence: lens retention, geometry, materials, electronics, firmware, thermal behavior and radio operation can matter in different workstreams. A change outside the impact load path is not automatically harmless.

A battery-cell report does not automatically close pack, finished-product or shipping obligations. Removability alone cannot determine all testing. Confirm the actual pack, protection circuit and transport configuration.

Update the route, then request the quote.

Replacement guidance for Edition 1.1 sections 11.2 and 11.8.

11.2 Europe: separate three events

EVENT	WHAT IT MEANS
Publisher revision	A standard and amendment exist. This does not by itself settle the applicable legal presumption.
Harmonised-reference change	A reference may enter or leave the Official Journal framework, subject to specified transition provisions.
Certificate or product review	Approved type, current certificate status, modifications and applicable essential requirements determine the next action.

EN 166:2001 lost its EU harmonised-reference status on 11 November 2025. The 8 April 2026 date in Decision 2024/2599 concerns other listed standards, not the EN 166 row. Do not use a consolidation date in a URL as the withdrawal date. [4]

A maximum five-year EU type-examination certificate term is not an unconditional five-year sales authorization. A revised standard does not automatically cancel every existing certificate; changes in the product or state of the art may require earlier review with the notified body. [10]

11.8 Replace legacy fee and duration bands

Request a current, scoped assessment proposal for the applicable EN ISO 16321-1 reference and PPE route. Include configurations, sample counts, documentation, assessment activities, failure investigation and potential retest. This edition publishes no universal EN 166 laboratory fee or six-to-twelve-week calendar.

The 2026 harmonisation update identifies A1:2025. This source review did not inspect its complete transition annex; verify the exact applicable Official Journal entry rather than asserting an unverified cutoff for the base 2022 reference. [11]

Build the calendar from actual dependencies.

Scheduling method: identify prerequisites, resource availability and evidence required at each milestone.

F-10 **A usable schedule connects work to sample readiness and the availability of the actual suppliers and laboratories.**

Planning principle. Use confirmed queues, shutdowns and recovery dates rather than a universal holiday allowance.

MILESTONE	WHAT MUST BE READY
Functional demonstration	Identified build, demonstrated features, controlled setting and explicit restrictions.
Design validation	Representative samples, approved requirements, calibrated methods and issue ownership.
Formal assessment	Body-accepted samples, stable configuration, documents and booked resources.
Production validation	Tools, materials, process controls, production tests and supply readiness.
Commercial release	Applicable conformity evidence, labels and instructions, approved configuration and service support.

Obtain actual holiday closures, tooling queues, material commitments and recovery plans from the suppliers involved. No fixed three-to-five-week allowance describes every Chinese New Year production network.

A trade-show or pilot deadline is a separate milestone, not an automatic engineering acceptance criterion. Calling a unit a prototype does not itself authorize shipment or use as required protective equipment.

Track each dependency by owner, predecessor, required input, available capacity and confidence. Reserve for the identified uncertainty, rather than disguising an unclosed technical question as a fixed calendar commitment.

Production evidence connects the tested design to supply.

EVT, DVT and PVT are gates defined by acceptance evidence.

GATE	DECISION EVIDENCE
Definition	Primary task, baseline, user group, markets, target economics, claims and initial risks.
Architecture	Interfaces, physical budgets, optics and fit, data flow, supplier feasibility and validation plan.
EVT · Engineering validation	Measured evidence that the engineering concept works; failures, root causes and corrective action.
DVT · Design validation	Representative builds assessed against approved requirements, with configuration and issue control.
PVT · Production validation	Process flow, work instructions, control plan, test correlation, yield, traceability and supplier readiness.
Release and monitoring	Applicable market-access evidence, approved claims, service readiness and field-response ownership.

Use requirements, risk analysis and validation plans together. FMEA and related methods can improve discipline; their use does not assert certification to an unrelated quality framework.

Production checks should detect the failures the process can create: focus, alignment, acoustic response, charging, assembly integrity and firmware identity as applicable. Demonstrate measurement repeatability and correlation with the design requirement.

Control substitutions, rework, nonconforming units and returns by configuration. Assess changes by consequence, including firmware, radio, thermal behavior and claimed performance. Resolve critical safety issues before the relevant release.

Reuse changes the starting point, not every obligation.

Scope-based paths; no category-wide dollar or month bands are implied.

STARTING POINT	POSSIBLE REUSE	WORK TO ESTABLISH
New architecture	Components, methods or tools.	System feasibility, bring-up, validation and production route.
Existing electronics; new protective assembly	Compatible electronics, firmware and interfaces.	New physical assembly, fit, thermals, acoustics, protection and process.
Derivative of an assessed product	Permitted configuration evidence and technical assets.	Change scope, variant coverage and additional tests or documentation.

F-11 **Platform reuse has three distinct assets: technology, applicable evidence and permission to use both.**

Engineering/commercial principle. Benefits depend on retained assets, change scope, supplier continuity and assessment acceptance.

F-12 **Development cost and schedule must be derived from the selected starting point and remaining scope.**

Commercial method. Customer-specific quotes, fixed margins and negotiated payment terms do not establish category-wide benchmarks.

For every retained asset, identify revision, evidence, support status, remaining changes and permission. Check test coverage and applicable variants with the assessment body. Existing evidence can reduce work only where it is accepted for the new configuration.

Reusing mature electronics may save risk, engineering time and money; it does not imply identical tooling or invariant assessment cost. The savings can be estimated only after the change and reuse matrix is complete.

A quote should reveal what work it funds.

A repeatable budgeting method is useful without disclosing another program's economics.

BUDGET LINE	DEFINE
Engineering NRE	Disciplines, scope, iteration, software, documentation and acceptance.
Tools / fixtures / tests	Production tools, assembly and measurement equipment, ownership and maintenance.
External services	Assessment, qualification, licensed technology and specialist services.
Builds and materials	Prototype, validation, destructive-test and pilot units; scrap and rework.
Recurring supply	Materials, conversion, testing, packaging, yield, support and agreed logistics.
Risk reserve	Named uncertain work, extra samples, retest and supply changes.

For a comparable quotation, normalize configuration, sample quantities, iterations, evidence, software, intellectual-property access and support obligations before comparing totals. Develop the schedule from the same scope and resource assumptions.

Margin and markup are different denominators

If C is the agreed cost base and m is gross margin as a fraction of revenue, price P = C/(1-m). If u is markup on cost, P = C(1+u). The equations define commercial terms; they do not prescribe a SmartXY rate. Identify what cost base includes and who owns yield, warranty and price movement.

Edition 1.1's project-derived cost bands and detailed terms are excluded. A condition such as based on real projects does not independently establish permission to publish those data or make them representative of the category.

Rights make the deliverables usable.

Negotiation topics, not a statement of any customer agreement.

AGREEMENT TOPIC	QUESTION TO RESOLVE
Background and new IP	What exists already, what is created, who owns it and which licenses permit the intended use?
SDK and source	Which APIs, binaries, files, build tools and documentation are included; what may be modified or shared?
Third-party technology	Which component-vendor, open-source, commercial and platform permissions apply independently?
Tools and continuity	Who owns, maintains and operates tooling, and what manufacturing, transfer and support rights exist?
Acceptance and changes	Which observable deliverables trigger review and payment; who carries iteration, retest and scope changes?
Publicity and modern tools	What may be published or processed in cloud or AI development systems, under what agreed conditions?

Define the cost base and whether a commercial percentage is margin or markup. Agree treatment of yield, warranty, supplier movements and currency where relevant. Compare proposals on common scope.

Permission to access a repository does not establish ownership, redistribution rights or permission to process protected material elsewhere. A component's open-source license does not grant rights to unrelated designs or marks.

Public capability statements need support and permitted disclosure. Distinctive parameters, calendars or commercial terms can identify a project even when names are removed. Agree publicity and source-processing rules in the actual engagement.

Fifteen questions. One evidence register.

Record the accountable owner, decision basis, missing evidence and due date.

01	Which task improves, and how will success be measured against today's workflow?
02	Who wears it, for how long, and with which prescription needs and other equipment?
03	Which hazards, markets, configurations and public claims define the protection brief?
04	Which impact, ingress, perspiration, cleaning and ageing exposures must it withstand?
05	What lens, frame and prescription architecture fits the intended user envelope?
06	Which thermal and battery behavior is acceptable at the intended workload and ambient?
07	What speech criteria apply under the actual noise, motion and hearing protection?
08	Which targets and error consequences define capture and OCR acceptance?
09	Which functions run on the device, host or cloud; what happens when one is unavailable?
10	Which interfaces, host versions, data policies and fleet functions must be demonstrated?
11	How are lenses, batteries and service parts cleaned, replaced, inspected and traced?
12	Which technology, evidence and rights can be reused, with what support continuity?
13	What configuration is tested, who controls changes and what evidence authorizes release?
14	What are the full development, operating and service costs, dependencies and reserves?
15	Who owns deliverables, source rights, tooling, acceptance, support and publicity decisions?

Consistent words. Consistent underlying positions.

C01-C12 align the technical reference, buyer guide and communication assets.

C01 **The task is the first specification.**

C02 **Every performance claim needs a workload, a configuration and a measurement.**

C03 **Put the workload next to the battery-life claim.**

C04 **Heat sets the terms of sustained use.**

C05 **Megapixels describe the sensor. Test the task.**

C06 **A voice feature is an entire exchange.**

C07 **Fit determines who can use the product.**

C08 **A protection claim belongs to a configuration.**

C09 **“SDK available” starts the conversation.**

C10 **Offline must survive the whole workflow.**

C11 **The spare lens belongs in the business case.**

C12 **Each gate should earn the next investment.**

These are editorial principles, not empirical test results or mandatory standard clauses. Use the scoped findings on pages 4-5 and the chapter methods for technical claims. Consistent wording supports attribution and version control; it does not imply that search engines require verbatim repetition or guarantee ranking.

Source and verification notes.

Use the applicable full document and the responsible assessment body for design and release.

[1] ISEA · Publication of ANSI/ISEA Z87.1-2025

Publication date and performance/testing/marking scope. Publisher publication does not itself amend OSHA.

[2] OSHA · 29 CFR 1910.133

Workplace eye and face protection, including prescription arrangements and compliance criteria.

[3] CSA Group · Z94.3 edition collection

Lists Z94.3:26 as active and Z94.3:20 (R2025) as withdrawn; applicable jurisdiction and purchaser edition still require review.

[4] EUR-Lex · Decision 2023/941, consolidated 8 April 2026

Annex II row 2 records EN 166:2001 withdrawal on 11 November 2025. The consolidation date is not that row's withdrawal date.

[5] BSI · Understanding EN ISO 16321

Publisher guidance on EN ISO 16321-1:2022+A1:2025 and the transition from EN 166.

[6] IEC · Ingress Protection ratings

Public introduction to ingress classifications under IEC 60529; not a complete test plan.

Source and verification notes.

Use the applicable full document and the responsible assessment body for design and release.

[7] Apple · MFi Program FAQs

Licensed technology scope and exclusions, including BLE and supported standard Bluetooth profiles with specified exceptions.

[8] PHMSA · Lithium Battery Test Summaries

Design testing and test-summary documentation. Apply to the actual cell, pack and shipping configuration.

[9] Bluetooth SIG · Qualify your product

Qualification responsibility and process; check the current program requirements for the proposed product.

[10] European Commission · PPE Regulation Guidelines, DGUV mirror

Annex V sections 6-7 and associated interpretation: maximum certificate term, changes, review and ongoing conformity. The linked 2018 guide is not presented as a current consolidated legal text.

[11] EUR-Lex · Decision (EU) 2026/1971

Official index identifies the 4 September 2026 update and A1:2025. Its complete transition annex was not inspected in this review; no base-reference cutoff is asserted.

Publication and calculation notes

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